Hello!

Pieter De Vis

Sam Hunt

Maddie Stedman

Samantha Malone

Joe Riordan

Today's Exercises

comet-toolkit.org/user-guide/training/vhroda

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National Physical Laboratory

Today's Agenda

1. **Presentation: CoMet Toolkit Introduction**
   - ❑ *What is CoMet?*
   - ❑ *Uncertainty 101*
   - ❑ *Tools Intro*
2. **Exercises**
   - ❑ **punpy** basics with in-situ type data
   - ❑ **obsarray** basics with EO type data

Today's Exercises

comet-toolkit.org/user-guide/training/vhroda

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The CoMet Toolkit – Uncertainties made easy

Pieter De Vis, Sam Hunt, Maddie Stedman, Samantha Malone, Joe Riordan
National Physical Laboratory

VHRODA hands-on tutorial - 19/11/2025

CoMet Toolkit

Python Tools

Uncertainty Handling

Open Source

Tested

HYPERNETS

esa truths

Applied

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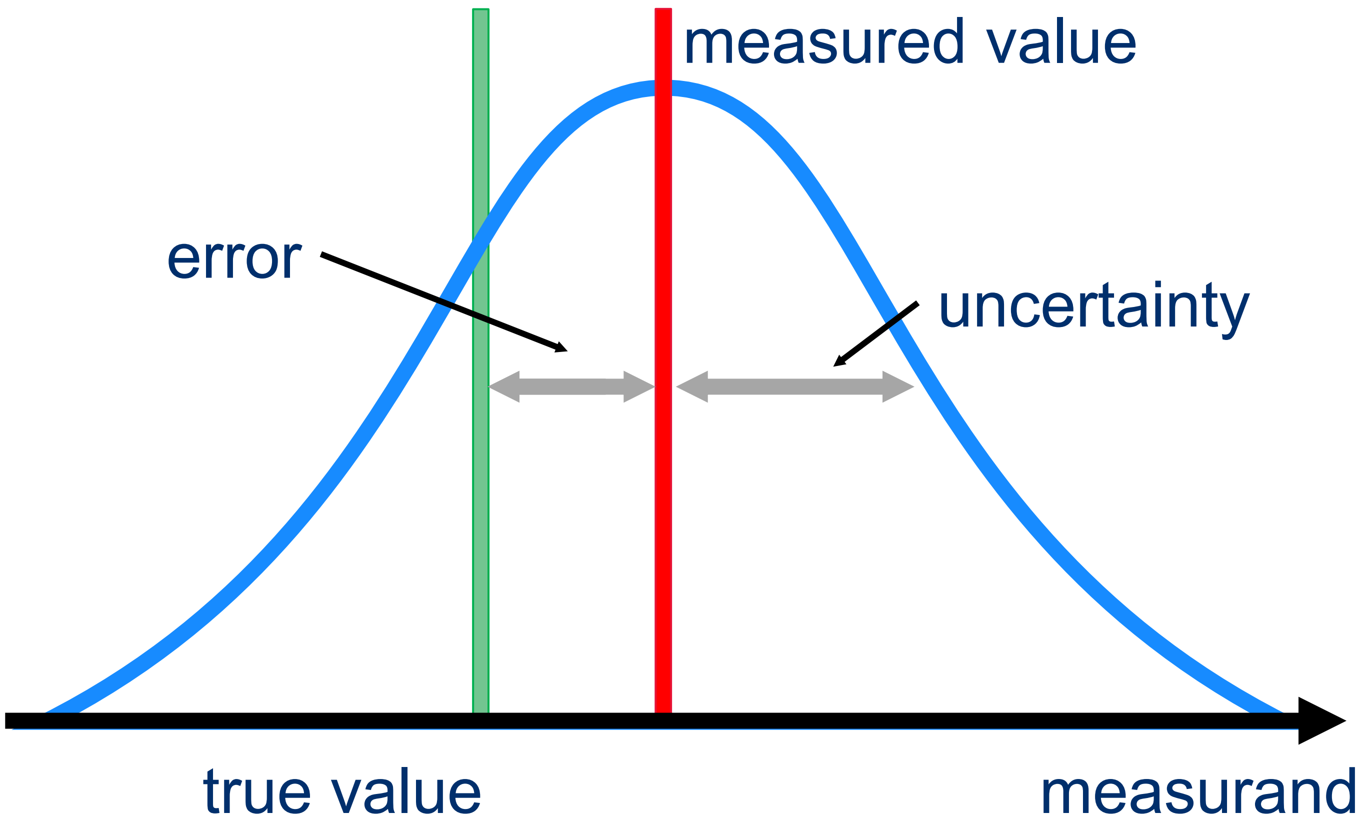
Uncertainties 101



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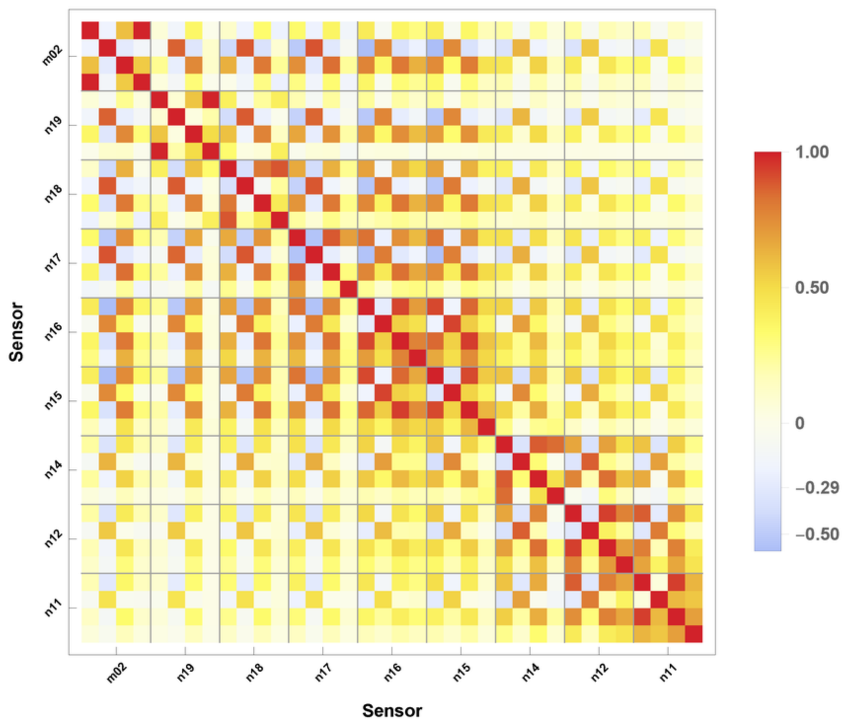
Error Correlation

What is Error Correlation?

Errors in a dataset are **not always independent** — e.g., when errors in pixels, bands instruments, or time steps are systematically related.

Why does it Matter?

- ❑ Bias uncertainties persist – averaging doesn’t help!
- ❑ Band ratios might be off – affecting retrieval uncertainty
- ❑ Misleading confidence in trends



Error correlation matrix from Giering et al. 2019

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Error Covariance

What is Error Covariance?

Combines error correlation and uncertainty

$$\boldsymbol{S} = \boldsymbol{U}\,\boldsymbol{R}\,\boldsymbol{U}^T$$

Random, systematic and structured uncertainty

- ❑ It is the errors that are correlated, not the uncertainty values
- ❑ Random means completely uncorrelated, i.e. error correlation is identity matrix
- ❑ Systematic means fully correlated, i.e. error correlation filled with 1's

A Metrological Approach

Uncertainties are evaluated and expressed following ***QA4EO Five Steps***, a framework which employs the principles of metrology.

MEASURAND 1 → **TRACEABILITY 2** → **UNCERTAINTY 3** → **CALCULATE 4** → **STORE 5**

1. **MEASURAND**: Define the measurand and measurement function
2. **TRACEABILITY**: Establish the traceability with a diagram
3. **UNCERTAINTY**: Evaluate each source of uncertainty and fill in an effects table
4. **CALCULATE**: Calculate the data product and uncertainties
5. **STORE**: Record information for users

CoMet Toolkit

punpy

Propagation UNcertainties in Python

obsarray

Handling uncertainty and error-covariance in datasets

comet_maths

CoMet mathematical algorithms and interpolation tools

UNC Specification

Uncertainty metadata naming conventions

CoMet Packages:

punpy

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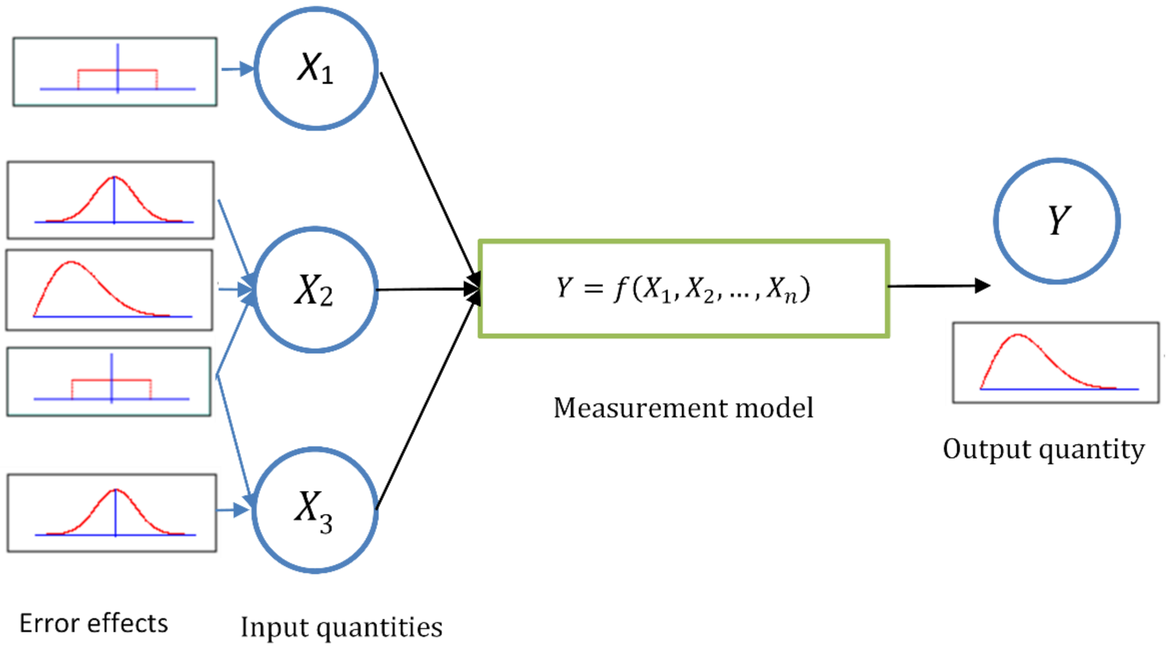
Propagating Uncertainties with punpy

- Python module for propagating **random**, **systematic** and **structured uncertainties** through any Python measurement function

- Flexible in terms of the **specified correlations** along given dimensions or between input quantities

- ***Monte Carlo*** and ***Law of Propagation of uncertainties*** methods available



Punpy as a Standalone Tool

punpy

- Simple user **interface**:
  1. Import punpy
  2. Define measurement function
  3. Create MC or LPU object
  4. Propagate uncertainties

```
import punpy
prop=punpy.MCPropagation(10000)
unc_measurand=prop.propagate_random(measurement_func,
      [input_qt1,input_qt2],[unc_qt1,unc_qt2])
```

- **Measurement function** are defined as python functions that take arrays as input quantities and return an array as measurand

- Many optional **keywords** for flexible functionality

  - *return_corr*
  - *Corr_between*
  - *Repeat_dims*
  - *Parallel_cores*
  - *Output_vars*
  - …

Punpy with digital effects tables

- punpy interfaces with obsarray to make uncertainty propagation as efficient and easy to use as possible
- *propagate_ds()* function returns an obsarray dataset with combined random, systematic and structured uncertainties on measurand

```
from punpy import MeasurementFunction

# Define your measurement function inside a subclass of MeasurementFunction
class IdealGasLaw(MeasurementFunction):
    def meas_function(self, pres, temp, n):
        return (n *temp * 8.134)/pres

# create object of the measurement function class and specify the variable names
gl = IdealGasLaw(["pressure", "temperature", "n_moles"], "volume", yunit="m^3")

# propagate uncertainties on the input quantities in ds to measurand in ds_y
ds_y = gl.propagate_ds(ds)
```

CoMet Packages:

obsarray

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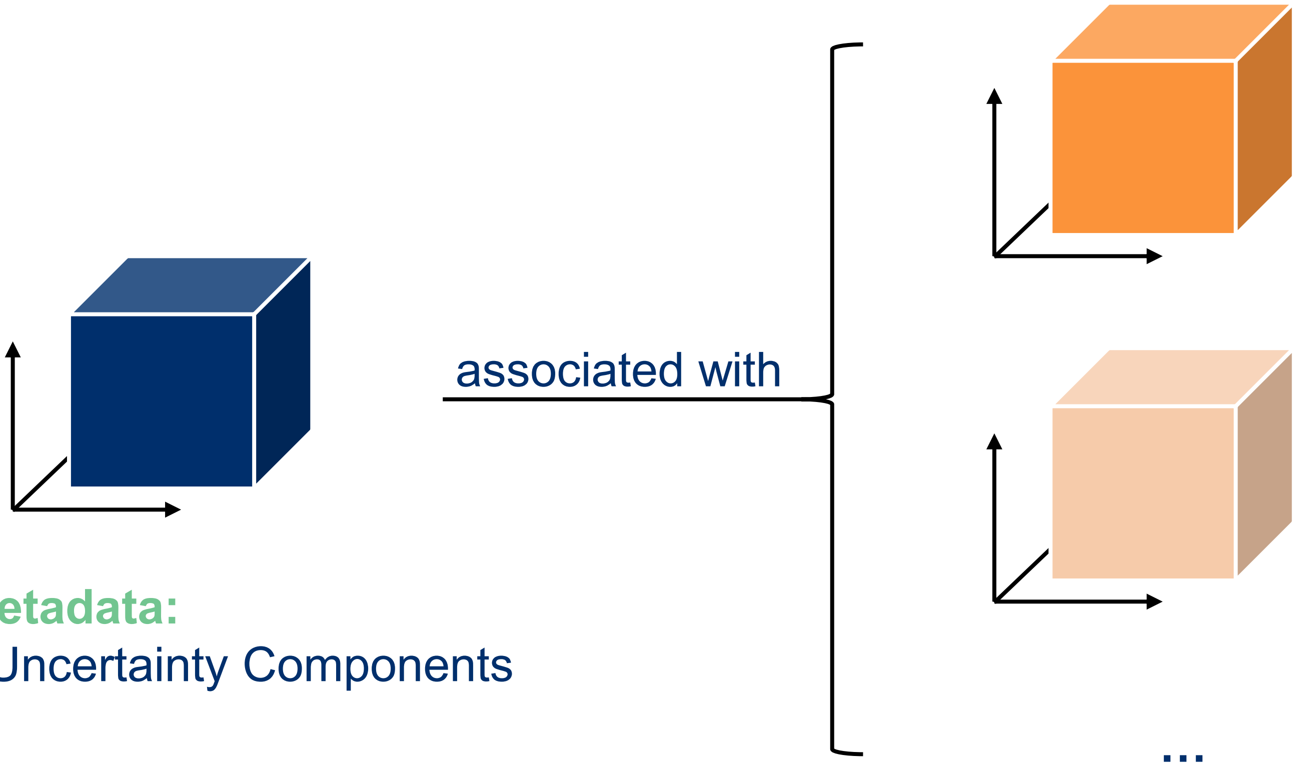
UNC Specification

obsarray

Uncertainty Variable Metadata

Observation Variables

Uncertainty Variables



Metadata:

- Uncertainty Components

Metadata:

- **PDF Shape** (gaussian, …)
- **Units** (abs. or rel.)
- **Error-Correlation**…

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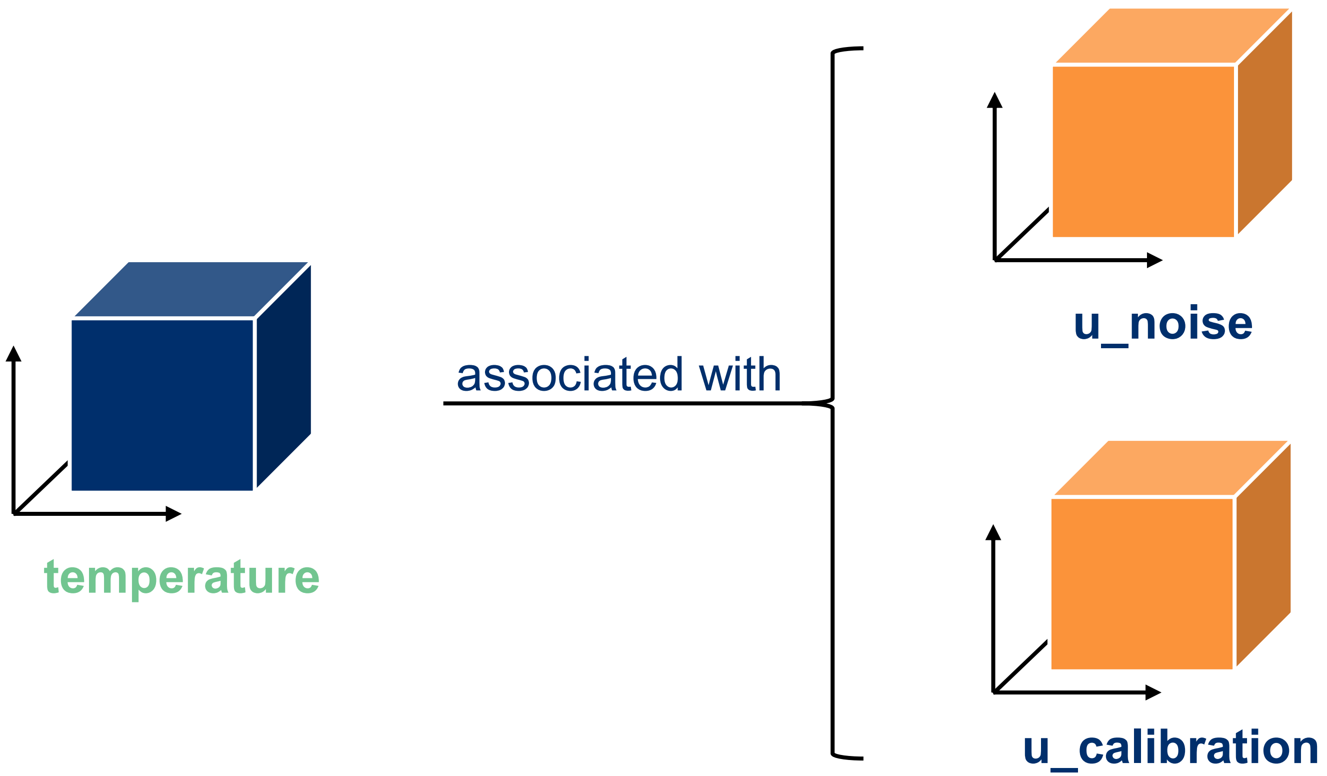
UNC Specification

obsarray

Example – Temperature Dataset

Observation Variables

Uncertainty Variables



Metadata:
- PDF Shape – “gaussian”
- Units – %
- Error-Correlation:
  - All dims - Random

Metadata:
- PDF Shape – “rectangular”
- Units – “K”
- Error-Correlation:
  - **x, y** – systematic
  - **time** – defined by matrix

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UNC Specification

Example – Temperature Dataset

```
variables:
  float temperature(time, lat, lon);
    temperature:unc_comps=["u_calibration", "u_noise"];
    temperature:units="K"
  float u_calibration(time, lat, lon);
    u_calibration:units="K";
    u_calibration:pdf_shape="rectangular";
    u_calibration:err_corr_dim1_name=["lat", "lon"];
    u_calibration:err_corr_dim1_form="systematic";
    u_calibration:err_corr_dim2_name="time";
    u_calibration:err_corr_dim2_form="err_corr_matrix";
    u_calibration:err_corr_dim2_params=["err_corr_calibration_time"];
  float u_noise(time, lat, lon);
    u_calibration:err_corr_dim1_name=["time", "lat", "lon"];
    u_calibration:err_corr_dim1_form="random";
  float err_corr_calibration_time(time, time);
```

obsarray

Measurement data handling in Python

- obsarray is an extension to **xarray** to support defining, storing and interfacing with measurement data – using the UNC specification.
- Also has functionality for defining flags following **CF Conventions**.
- Working towards
- It is designed to work well with **netCDF** files and for the **Earth Observation** community.

Plugs straight into punpy for propagation through measurement functions!

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CoMet Application Examples

punpy

obsarray

comet_maths

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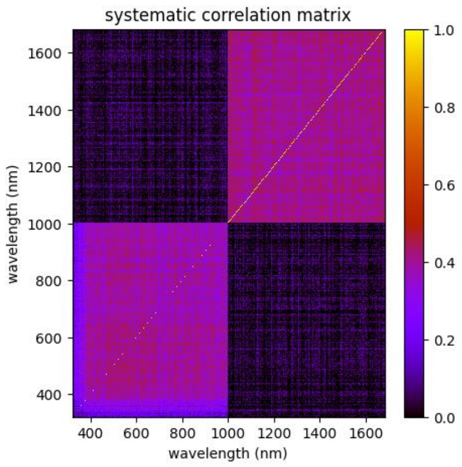
CoMet Toolkit in Action

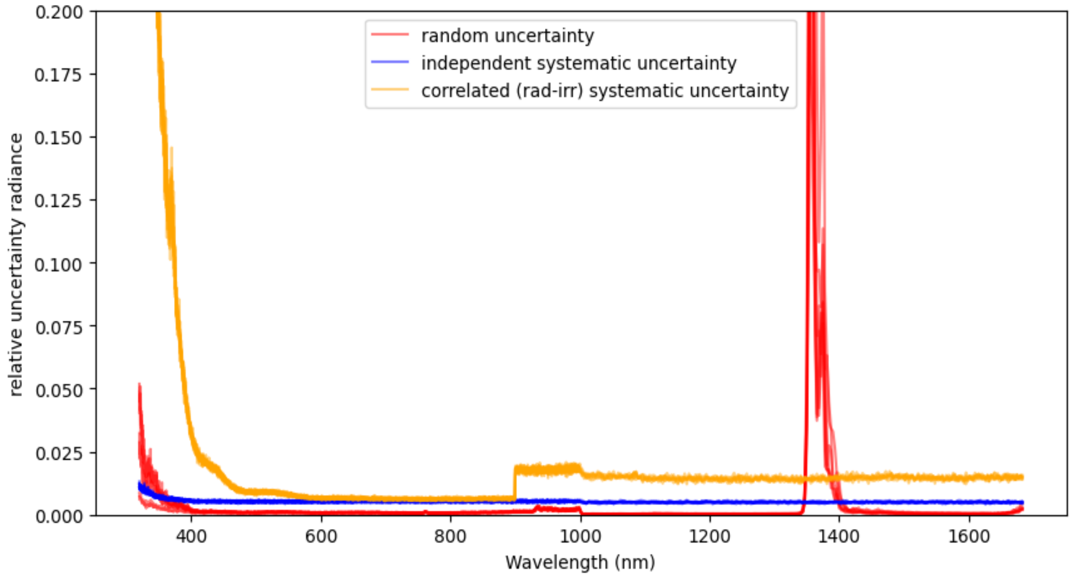
- Validated against **NIST** uncertainty engine:
  https://colab.research.google.com/github/comet-toolkit/comet_training/blob/main/NIST_example.ipynb

- *CoMet* is used in various other projects, such as **QA4EO**, **HYPERNETS**, **CHIME L2**, **FLEX** validation, **TRUTHS** science studies, **LIME**, **FRM4SOC**, **RPV4PICS**
  see https://www.comet-toolkit.org/projects/

- **Example** from *hypernets_processor* :

  Hypernets is an automated network of in-situ instruments measuring reflectance for L2 satellite validation





Exercises

- ❑ Please go to www.comet-toolkit.org/user-guide/training/

Today's Exercises

comet-toolkit.org/user-guide/training/vhroda

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🎉 CoMet Release

- V1.0 of Comet toolkit has been released as **open source** toolkit:
  - www.comet-toolkit.org
  - github.com/comet-toolkit
- Accompanied by training material (**Jupyter** notebooks hosted on google colab):
  - www.comet-toolkit.org/examples
- Documentation & ATBD for individual tools:
  - obsarray.readthedocs.io/en/latest/
  - punpy.readthedocs.io/en/latest/
  - comet-maths.readthedocs.io/en/latest/

Outlook

- ❑ **Current release** will be presented De Vis & Hunt (in prep)

- ❑ Looking to continue to expand the use cases the developed tools
  - Aiming to enable uncertainty propagation through **any python measurement function**
  - Please get in touch if you are interested!

- ❑ This has been our first step into this way of working, **many more ideas in a roadmap**
  - MetEOR (e.g. CEOS-PVP Cal/Val pipeline, BRDF tool, …)

- ❑ **1.5 day training course** tentatively planned for May at NPL

MetEOR

www.comet-toolkit.org/meteor/

https://qa4eo.org/met4eo/

Summary

- The **CoMet toolkit** is an open-source software project to develop Python tools for the handling of error-covariance information in the analysis of measurement data

- This toolkit is based on **robust metrology**, and makes dealing with complexities of uncertainties much easier

- Includes **obsarray**, **punpy** **&** **comet_maths** as initial offering, to be extended

- These tools are already being used operationally in various projects (e.g. Hypernets)